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Magnetostratigraphy of the Willwood Formation, Bighorn Basin, Wyoming: new constraints on the location of Paleocene/Eocene boundary

L. Tauxe [a,b], J. Gee [c], Y. Gallet [d], T. Pick [a,b], T. Bown [e]

[a] *Scripps Institution of Oceanography, La Jolla, CA, USA*
[b] *Fort Hoofddijk Paleomagnetic Laboratory, The University of Utrecht, Fort Hoofddijk, Budapestlaan 17, 3584 CD Utrecht, The Netherlands*
[c] *Lamont Doherty Earth Observatory, Palisades, N.Y., USA*
[d] *Institut de Physique du Globe, Université de Paris, Paris, France*
[e] *U.S. Geological Survey, Denver, CO, USA*

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Abstract

The lower Eocene Willwood Formation in the Bighorn Basin of Wyoming preserves a rich and diverse mammalian and floral record. The paleomagnetic behavior of the sequence of floodplain paleosols of varying degrees of maturation ranges from excellent to poor. We present a magnetostratigraphic section for a composite section near Worland, Wyoming, by using a set of strict criteria for interpreting the step-wise alternating field and thermal demagnetization data of 266 samples from 90 sites throughout the composite section. Correlation to the geomagnetic reversal time scale was achieved by combining magnetostratigraphic and biostratigraphic data from this section, from a section in the Clark's Fork Basin in northern Wyoming, and from DSDP Site 550, with the isotopic date determined on a tuff near the top of our section. Our correlation suggests that the Bighorn Basin composite section in the Worland area spans from within Chron C24r to near the top of Chron C24n, or from approximately 55 to 52 Ma. This correlation places the Paleocene/Eocene boundary within the vicinity of the base of the section. Cryptochron C24r.6 of Cande and Kent is tentatively identified some 100 m above the base of the section. The temporal framework provided here enables correlation of the mammalian biostratigraphy of the Bighorn Basin to other continental sequences as well as to marine records. It also provides independent chronological information for the calculation of sediment accumulation rates to constrain soil maturation rates. We exclude an age as young as 53 Ma for the Paleocene/Eocene boundary and support older ages, as recommended in recent time scales. The location of a tuff dated at 52.8 ± 0.3 Ma at the older boundary C24n.1 is consistent with the age of 52.5 Ma estimated by Cande and Kent and inconsistent with that of 53.7 Ma, from Harland et al.

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1. Introduction

The Willwood Formation in the Bighorn Basin of Wyoming (Fig. 1) contains a rich record of early Eocene floral and mammalian evolution [3–5]. Bown et al. [6] compiled the fossil localities of the Bighorn Basin near Worland, Wyoming, into a composite section. Willwood deposits consist of moderately mature and mature paleosols developed on overbank deposits [7]. These alternate vertically with crevasse-splay complexes that consist of immature paleosols and pedogenically unmodified siltstones and sandstones [8]. Sheet sandstones, representing the trunk channels, locally truncate these flood deposits. The formation is characterized by colorful paleosols described in detail by [7].

The age of this, some 700 m thick, composite section was constrained by Wing et al. [4], who published an ^{39}Ar–^{40}Ar date of 52.8 ± 0.3 Ma on a tuff near the top of the Willwood (634 m in the composite section). They quoted a correlation of a floral datum occurring just below the base of the Willwood (the FAD of *Platycarya* pollen at −35 m in the composite section) to the NP9/NP10 nannofossil zone boundary in the Gulf coast [9]. The NP9/NP10 boundary is often used for recognizing the Paleocene/Eocene boundary in marine sections [see 10 for a thorough discussion of the problem of recognizing the Paleocene/

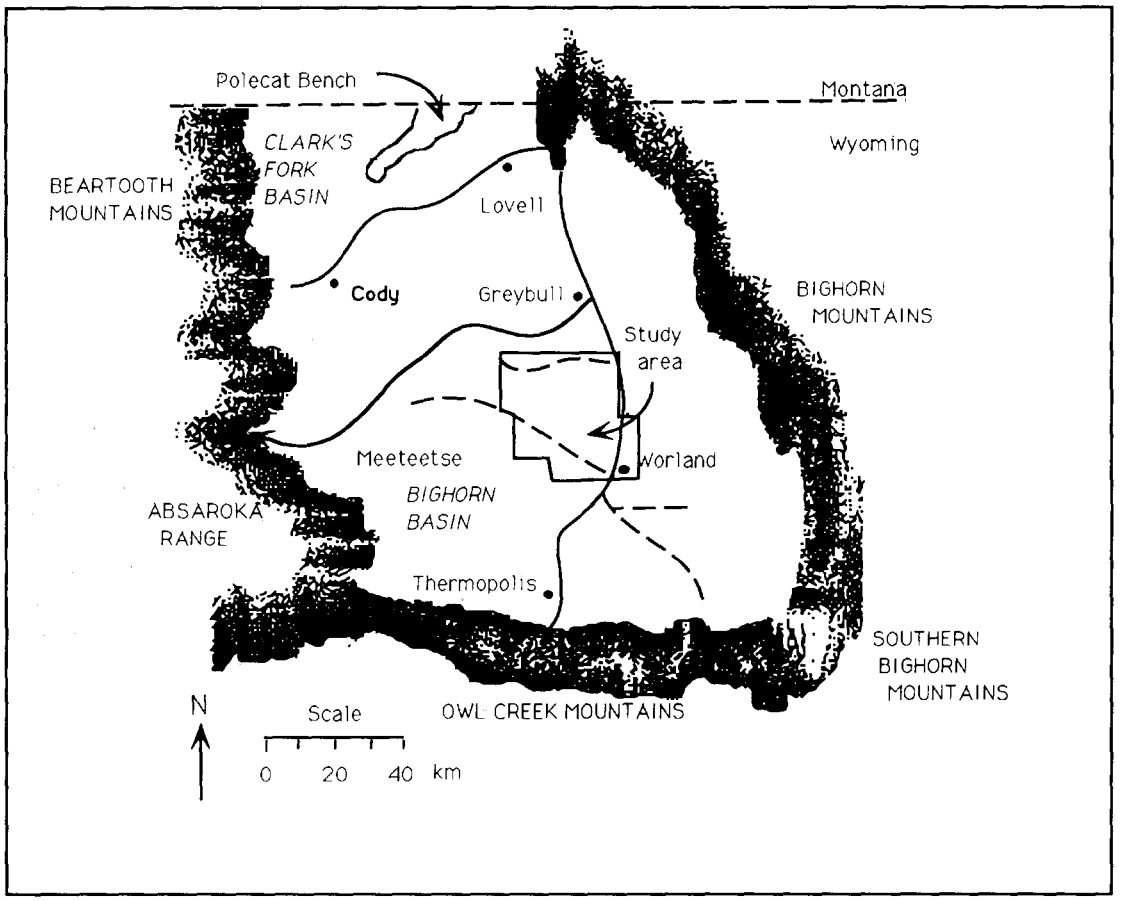


Fig. 1. Map showing location of study area (108.5°W, 44.2°N). Also noted is the location of the Polecat Bench in the Clark's Fork Basin, where Butler et al. [25] took the samples mentioned in the text.

Eocene boundary]. Wing et al. [4] assumed an age of 55.7 Ma for the NP9/NP10 boundary [11], although age estimates for this boundary range from as young as 53 Ma to nearly 57 Ma [see 1 and 2 for discussion of the calibration of the Paleogene time scale]. Apart from these tie points, internal temporal control has been estimated using a relative scale based on assumed pedogenic maturation rates [12]. Since soil maturation rates are a function not only of time but also of climate, independent time control is necessary to validate the model of sediment accumulation rate derived from paleosol descriptions. Furthermore, equivalence of the FAD of *Platycarya* with the Paleocene/Eocene boundary is not well established [see 9] and its actual position within the composite section is poorly documented [compare 6 and 4, for example]. Thus, the temporal framework for the interpretation of floral, faunal, sedimentological and isotopic information from this section, and co-eval sections in the Clark's Fork Basin to the north, would greatly benefit from additional time constraints, such as a magnetostratigraphic pattern.

It is the purpose of this paper to present new magnetostratigraphic data for the Willwood Formation in the Bighorn Basin. These data provide a means of correlating the sediments and included floras and faunas to the geomagnetic reversal time scale (GRTS). Our correlation sheds light on the possible location of the Paleocene/Eocene boundary in the Bighorn Basin composite section, as well as in the contemporaneous and older sediments exposed to the north. The magnetostratigraphic interpretation allows estimation of sediment accumulation rates, which provide independent constraints on soil maturation rates. Finally, the data presented here tie the isotopic date from the top of the Willwood Formation directly to the GRTS, establishing a much needed calibration point for Chron C24n.

2. Sampling and paleomagnetic analysis

Three to five separately oriented hand samples were collected from 90 paleomagnetic sampling sites spanning the entire Willwood composite section and extending some 50 m into the overlying Tatman Formation. Apart from sandy lithologies, which were avoided, an effort was made to sample every kind of silt and clay lithology present. Sampling sites are referenced with respect to the composite section of Bown et al. [6]. Samples were sanded into nominal cubic-inch specimens and analyzed in the Scripps paleomagnetic laboratory. One specimen from each of 43 sites was subjected to stepwise thermal demagnetization from 120°C up to the maximum blocking temperature. The magnetic behavior ranged from excellent to poor. All remaining specimens were treated first by step-wise alternating field demagnetization, from 1 mT up to a maximum of 80 mT, followed by thermal demagnetization in steps from 100°C to the maximum unblocking temperature. The range of behavior is illustrated in Fig. 2. A few specimens had maximum unblocking temperatures above 600°C, suggesting a hematite remanence, but the majority were completely demagnetized by 600°C and some even by 200°C. The examples in Figs. 2a and b turn toward the origin at about the 200°C step and are demagnetized by about 550°C, suggesting a magnetite carrier. Other samples have vectors that first 'hover' and then become random before closing in to the origin (Fig. 2c), or have no coherent behavior at all (Fig. 2d). Many specimens have demagnetization trajectories best described by a great circle; in these cases a characteristic component was not isolated, only the plane in which it lies (Fig. 2e,f).

Because of the great variety of results, we adopted the following approach. Specimens similar to those shown in Fig. 2a and b were treated using principal component analysis [13]. The best fit line was used if it was defined by at least four consecutive demagnetization steps and had a maximum angle of deviation [MAD; 13] of less than 25. Specimens similar to that shown in Fig. 2c were treated by calculating a Fisher mean [14] of a minimum of three consecutive demagnetization steps and used if the data had a circular standard deviation [CSD; 14] of less than 35. Specimens such as that shown in Fig. 2e, were analyzed by great circles and used if they had a MAD less than 25. Data from specimens that were completely demagnetized by 250°C, showed

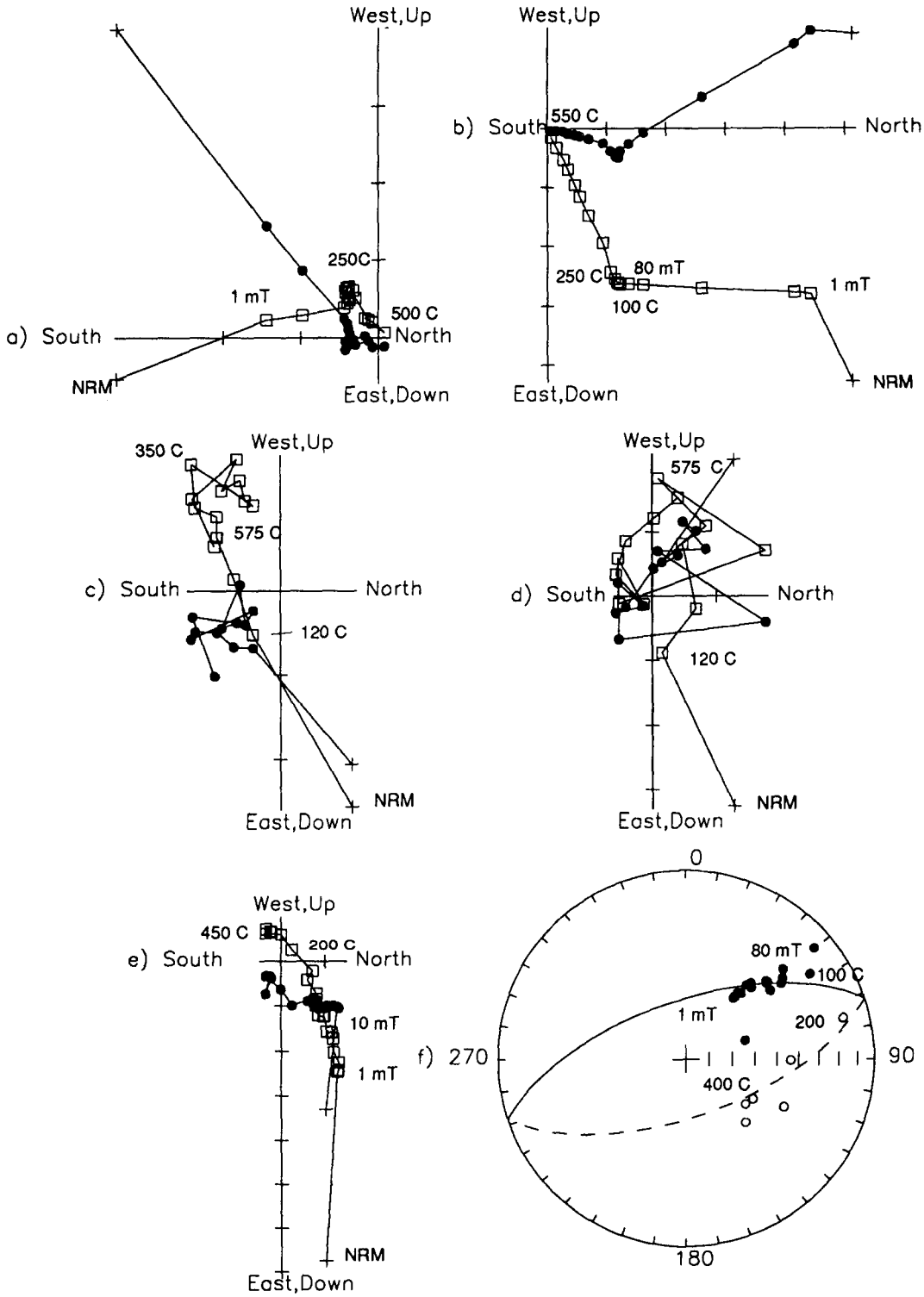


Fig. 2. (a–e) Vector end-point diagrams of representative behavior during step-wise demagnetization. Solid symbols = the horizontal plane; open symbols = the east–west vertical plane. (a) and (b) are suitable for principal component analysis [13]; (c) is more amenable to calculation of a Fisher mean [14]; (d) cannot be interpreted; (e) is best fit by a great circle, as shown in (f). (f) Equal area projection of data shown in (e). Solid symbols = the lower hemisphere; open symbols = upper hemisphere.

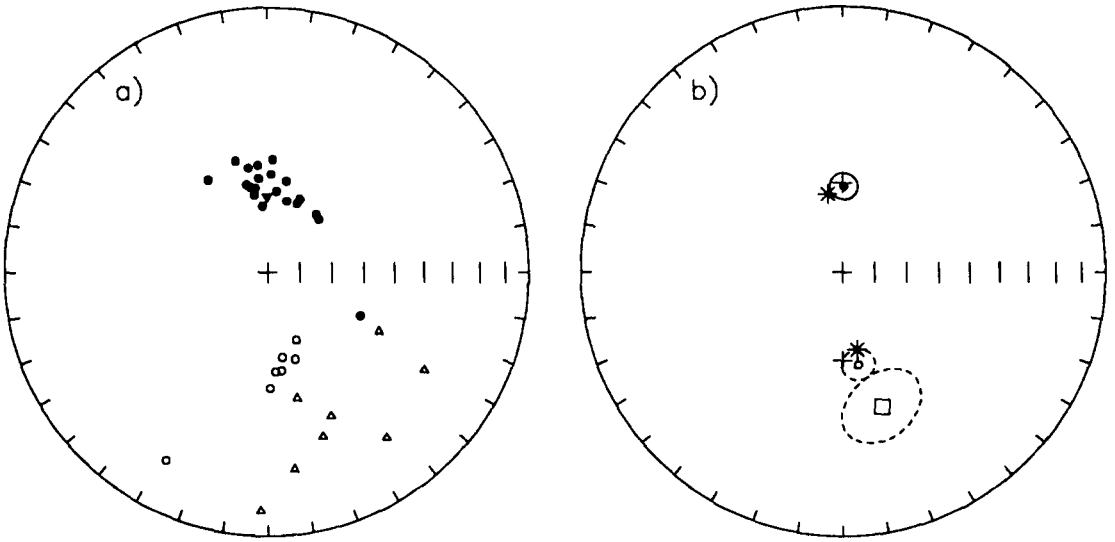


Fig. 3. Equal area projections (see caption for Fig. 2e). (a) α (dots and circles) and β (triangles) site means. (b) Means of normal (dot) and reversed modes (square for α plus β sites and circle for α sites only, excluding the single direction with shallow inclination as described in text). + = present dipole field; * = the expected Eocene direction (calculated from the paleomagnetic pole of [17]) and has a (normal) declination of 349° and an inclination of 66°.

random demagnetization behavior (Fig. 2d), or otherwise failed to meet any of the foregoing criteria were ignored.

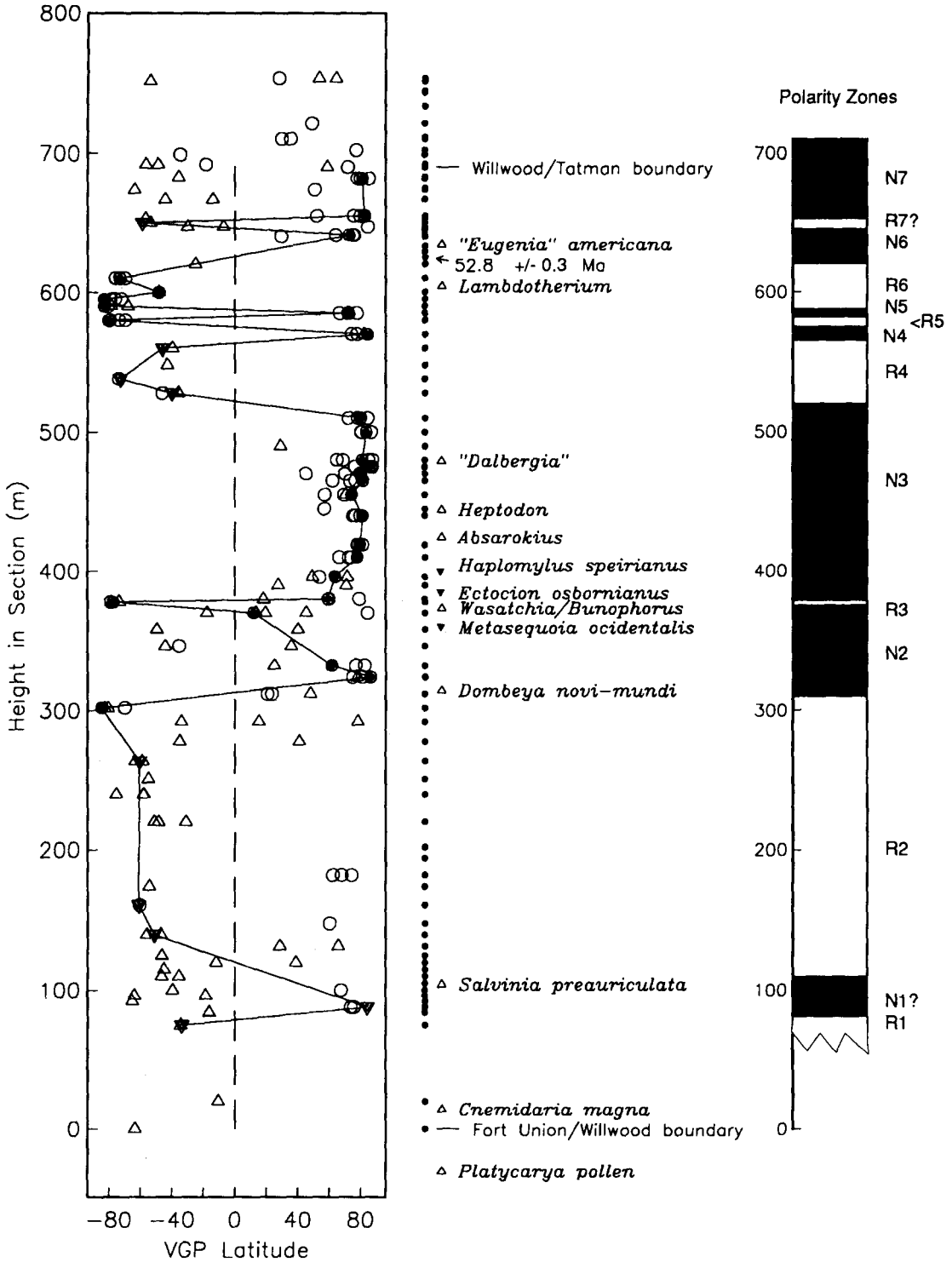
Site means were calculated in two ways. Means for sites having only directional data were calculated in the usual way, whereby each direction was assigned unit weight and a vector mean and CSD were estimated using Fisher statistics [14]. Sites whose dispersions exceeded the cut-off value for a random distribution defined by Watson [15] (CSD ~ 35°C for $N = 3$ specimens) were not used; those surviving were designated α sites and constitute our most reliable data. Sites including great circles were analyzed using the method of McFadden and McElhinny [16] for combining great circles and directed lines. Site means with estimated α_{95} values of less than 45° were designated β sites. Sites failing these two criteria were not considered further.

The natural remanent directions were highly scattered but had their highest concentration near the present dipole field direction (declination = 0°, inclination = 63°). The site mean directions of α and β sites are shown in Fig. 3a. These separate into two groups, one directed north and down (normal) and the other southeast and up (reversed). Mean directions and confidence ellipses were calculated in several ways (Table 1).

Table 1
Summary of paleomagnetic data

Mode	N	$\overline{Dec}$	$\overline{Inc}$	η_{95}	Dec_η	Inc_η	ξ_{95}	Dec_ξ	Inc_ξ
Normal (α only)	20	0.8	64.1		$\alpha_{95}=$			3.9	
Reversed ($\alpha+\beta$)	15	163.5	−46.0	13.3	296.8	−33.8	10.27	224.9	24.8
Reversed (α only)	7	179.3	−56.8	17.6	46.3	−24.8	1.4	305.9	−21.3
Reversed (α except one)	6	170.1	−61.3		$\alpha_{95}=$			5.0	

$\overline{Dec}$ and $\overline{Inc}$ are the estimated values for the true mean directions. Dec_η and Inc_η are the directions of the major semi-axes and Dec_ξ and Inc_ξ are for the minor semi-axes, whose semi-angles of 95% confidence are η_{95} and ξ_{95}, respectively. These are estimated by the bootstrap technique of [18].


Height in Section (m)
800
700
600
500
400
300
200
100
0
−80 −40 0 40 80
VGP Latitude
Willwood/Tatman boundary
"Eugenia" americana
52.8 +/- 0.3 Ma
Lambdotherium
"Dalbergia"
Heptodon
Absarokius
Haplomylus speirianus
Ectocion osbornianus
Wasatchia/Bunophorus
Metasequoia ocidentalis
Dombeya novi-mundi
Salvinia preauriculata
Cnemidaria magna
Fort Union/Willwood boundary
Platycarya pollen
Polarity Zones
700
600
500
400
300
200
100
0
N7
R7?
N6
R6
N5
<R5
N4
R4
N3
R3
N2
R2
N1?
R1

One site has a direction that is east–southeast and down, is intermediate between the normal and reversed directions, and was excluded from the calculation of mean directions in the following. The normal mode of directions is Fisher distributed and the mean and α_{95} are shown in Fig. 3b. The mean direction is indistinguishable from either the present dipole field (marked with a plus in Fig. 3b) or the expected Eocene direction calculated from the paleopole of Diehl et al. ([17]; marked with a star in Fig. 3b). The reversed mode calculated using both α and β sites is not Fisher distributed and the 95% confidence limits were estimated using the bootstrap method of Tauxe et al. [18] (see Table 1 and the open square in Fig. 3b). The combined data set is not antipodal to the normal mode, based on the bootstrap reversals test of [18]. However, using only α sites, the reversed mode is antipodal to the normal mode, though still not Fisher distributed. Excluding one southwesterly direction with a very shallow inclination, the remaining α sites are Fisher distributed and are antipodal to the normal mode (see circle in Fig. 3b). The mean calculated in this way is also indistinguishable from either the present reversed dipole field or the expected Eocene reversed direction. Unfortunately, there is no possibility of a fold test in these flat-lying beds.

It is possible that the site mean directions surviving our rather stringent acceptance criteria are entirely overprinted in recent fields and that they do not represent ancient magnetizations at all. If this is the case, then the data presented here cannot be used even for magnetostratigraphic purposes. However, because the normal characteristic directions are carried in most cases by what appears to be magnetite (Fig. 2b) and roughly half the reversed site means are calculated by great circle analysis (triangles, Fig. 3a), it is likely that the non-antipodality of the normal and reversed directions (using both α and β sites) is caused by unremoved normal overprints in the β sites which biases the combined reversed mean. It is worth noting that all but one of the normal sites come from the upper half of the section, which has suffered the least pedogenic alteration [12,19]. Furthermore, the normal data come from several different lithologies, representing a variety of pedogenic alterations. Under the reasonable interpretation that the data represent geomagnetic polarity from near the time of deposition, the data may be used for magnetostratigraphic purposes because the underlying polarity is not in doubt. Taking the latter view, then, we proceed with a discussion of the magnetostratigraphic consequences of the data.

3. Magnetostratigraphic results

The individual specimen directions, as well as the site mean directions, were converted to virtual geomagnetic pole positions (VGP's). The VGP latitudes are plotted in Fig. 4, with each type of data analysis denoted by a different symbol so that overall data quality may be readily assessed. The locations of all the paleomagnetic sampling sites are shown as dots to the right of the column, together with key index taxa [from 4]. Formational boundaries and the location of the tuff dated by [4] are also noted. Positive VGP latitudes are interpreted as normal sites and negative latitudes as reversed ones; the resulting polarity log is shown to the right. Polarity zones are labelled from R1, at the bottom, to N7, at the top. The two zones based on a single β site are marked with a '?' to indicate that they are less well established than those based on α sites or multiple sites.

Fig. 4. Left: Magnetostratigraphic results of southern Bighorn Basin composite section plotted as VGP latitudes versus height. ○ = principal components; △ = Fisher means; ● = type α site means (directions only); ▲ = β site means (includes great circle information, see text). Dots to right of paleomagnetic data are sampling sites. Middle: △ = first appearance of index taxa; ▼ = last appearance of index taxa [data from 4]. Formational boundaries are indicated by horizontal lines. The stratigraphic placement of the isotopic date of [4] is shown by an arrow. Right: interpreted polarity log. Polarity zones based on a single β site are indicated by question marks.

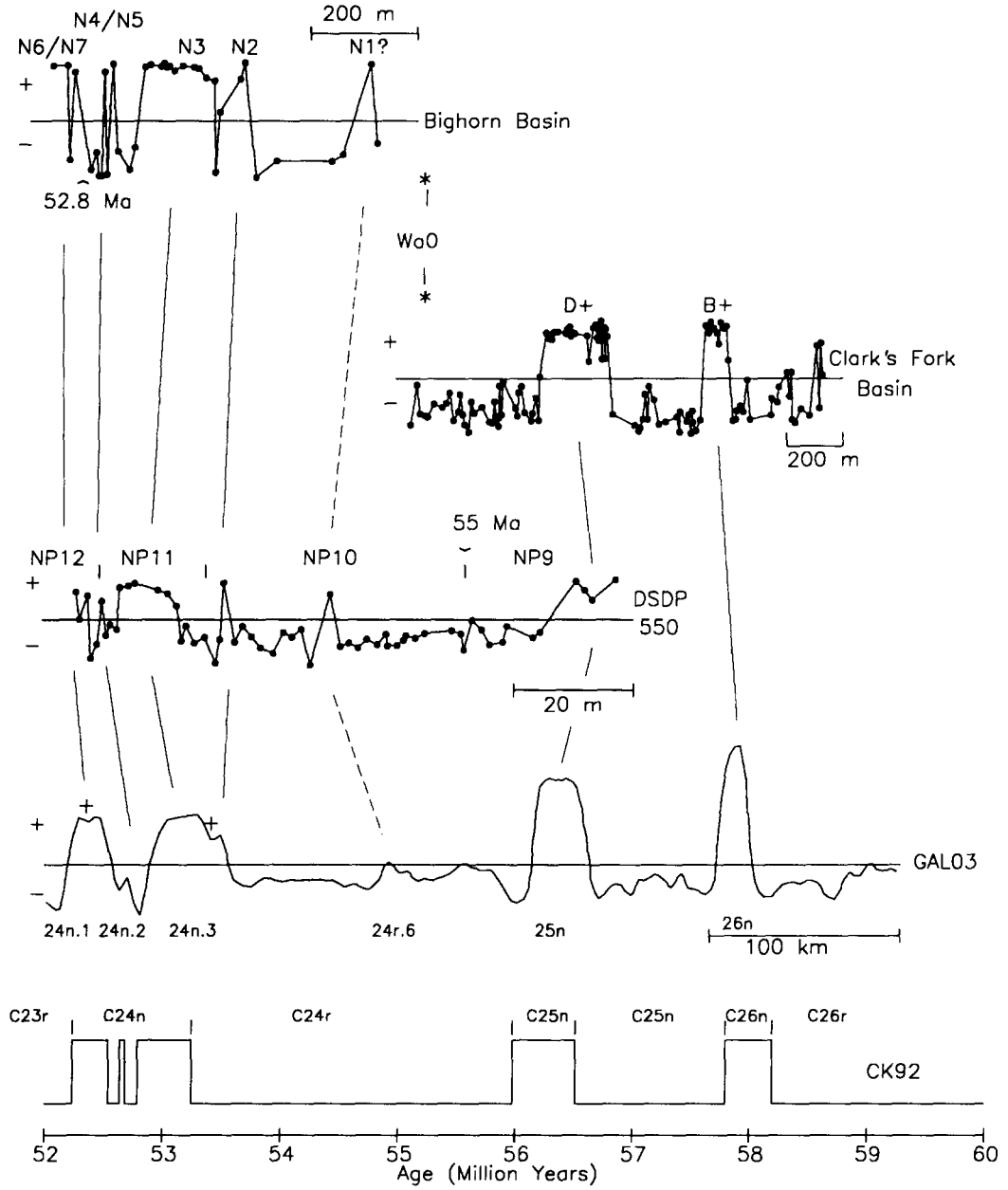


Fig. 5. Summary diagram showing the geomagnetic reversal time scale (GRTS) of Cande and Kent [1] (CK92) at the bottom. Above CK92 the magnetic anomaly profile from GAL03 [digitized from 1] is plotted, with certain features labelled, as referred to in text and in [1]. Above GAL03, we plot the inclination data from table 4 of [23] (excluding one point with anomalously high intensity). Nannofossil zone boundaries are from table 10 of [24]. Noted is the 55 Ma date from Swisher and Knox [unpublished and quoted in 1] at the NP9/NP10 boundary. Above the DSDP 550 data are VGP latitudes from the Clark's Fork Basin [digitized from 25]. Their B + and D + magnetozones are indicated, as well as the 1520 m level containing Wa0 mammal fossils described by [25]. At the top are the VGP latitudes from Fig. 4 (Bighorn Basin). Also shown is the location of Wa0 mammal fossils at 0 m and the level of the isotopic date of [4]. Solid lines = correlations to the GRTS; dashed line = a tentative correlation of various magnetozones to anomaly 24r.6.

4. Discussion

4.1. Correlation to the geomagnetic reversal time scale

Correlation of our data to the geomagnetic reversal time scale (GRTS) is subject to the constraints summarized in Fig. 5. The GRTS of Cande and Kent (CK92; [1]) is shown at the bottom. The polarity pattern is based on the stacking of several magnetic anomaly profiles, such as GAL03, shown above the square model of CK92. The age estimates of reversals in CK92 are based on a spline interpolation through several calibration points chosen throughout the Cenozoic. One such calibration point is the Paleocene/Eocene boundary. Cande and Kent [1] quote an age of 55 Ma from Swisher and Knox (unpublished) for the nannofossil zonal boundary NP9/NP10. This boundary is often used to identify the Paleocene/Eocene boundary in marine sections [see, e.g., 10 for an extensive discussion of the problem]. Perhaps the best magnetostratigraphic record for marine sediments in the Paleogene was obtained from the Contessa Highway Section in the Umbrian Region of Italy [20]. Unfortunately, the key taxa defining the NP10 nannofossil zone were not identified; hence the precise location of the NP9/NP10 boundary is not known. Paleogene sequences in the London Basin were also studied paleomagnetically by [21] and biostratigraphically by [22]. Although each magnetozone can be approximately correlated to the time scale, the sequences are not continuous and do not serve as a useful reference section. Other marine sections spanning the late Paleocene and early Eocene are from cores obtained through the Deep Sea Drilling Project (DSDP) and the Ocean Drilling Program (ODP). Apart from DSDP Hole 550, there appear to be either hiatuses, scattered paleomagnetic data, or difficulty in identifying the NP9/NP10 boundary. Therefore, we plot the inclination data from DSDP Hole 550 [23] (excluding one point with an anomalously high intensity) in Fig. 5. Nannofossil boundaries were located in table 10 of [24] and are also shown in Fig. 5. The NP9/NP10 boundary lies within a broad reversed zone readily correlated to C24r. Thus, the 55 Ma date can be tied to a point in the first trimester of C24r. CK92 followed [10] in assuming a position precisely at the end of the first trimester for their calibration point.

Within the Bighorn Basin composite section itself, there is an ash bed dated at 52.8 ± 0.3 Ma. We sampled through the sequence containing the ash and the date is thus tied to the magnetostratigraphic pattern, as shown in Figs. 4 and 5. In addition, there is a thick section of the Fort Union Formation (also known as the Polecat Bench Formation). This underlies the Willwood Formation, whose magnetostratigraphy along the Polecat Bench in the Clark's Fork Basin (Fig. 1) was investigated by Butler et al. [25] (Fig. 5). The two sections can be confidently correlated based on the mammalian faunal zonations. The earliest Wasatchian fauna, Wa0, occurs at the base of the Bighorn Basin composite section near Worland and at 1520 m in the Clark's Fork Basin [26] section, as shown by asterisks in Fig. 5.

Butler et al. [25] correlated their polarity zones B+ and D+ (Clark's Fork Basin) to magnetic polarity chrons C26n and C25n, respectively, as shown in Fig. 5. Taking the composite pattern of the two sections and the isotopic date from Wing et al. [4]. We tie R1/R2 to C24r, N2/N3 to C24n.3, N4/N5 to C24n.2 and N6/N7 to C24n.1 (solid lines in Fig. 5).

4.2. 'Tiny wiggles' and the 'Oldhaven magnetozone'

Cande and Kent [1] re-evaluated the entire Paleogene magnetic reversal sequence, including a thorough characterization of numerous 'tiny wiggles' noted throughout this time period. Tiny wiggles are very short features in the marine magnetic anomaly patterns that occur in several independent anomaly profiles. They can be modelled as very short polarity zones or as intensity fluctuations [see 27 for a more detailed discussion]. Between anomalies 24 and 25, for example, ten such short wavelength features were noted. This interval of the time scale was based on several profiles. One of these, GAL03 from the Indian Ocean, is shown in Fig. 5. Because of the nature of stacking, some short wavelength fea-

tures, such as 24n.2 appear as first-order polarity intervals in the time scale, whereas others (see examples marked by plus signs in GAL03) do not merit mention. Many, such as the tiny wiggle labelled 24r.6, are designated 'cryptochrons' in their recent time scale [1]. Thus, the actual polarity history of the geomagnetic field cannot be determined from a single anomaly profile, or indeed even from a stack of several. High resolution magnetostatigraphic sections are essential for delineating the shorter wavelength features of the polarity time scale.

In Fig. 5 we show not only a representative magnetic anomaly profile, but the VGP latitudes of our continental section from the Southern Bighorn Basin and the inclination data from DSDP Hole 550, as previously mentioned, with the preferred correlation to C24n shown as solid lines. It appears plausible that one of the tiny wiggles of GAL06, 24r.6, appears in both the Bighorn Basin composite section and DSDP 550 as well, as suggested by the dashed lines. The correlative cryptochron (C24r.6) spans the interval 54.49–54.5 Ma in CK92, some 0.5 Ma later than the assumed position of the Paleocene/Eocene boundary.

It is interesting to note that one of the magnetozones from the London Basin (the so-called, 'Oldhaven magnetozone') is a normal zone, obtained from sediments probably belonging to the NP10 nannofossil zone [see 22]. It is, therefore, possible that our N1?, the single point normal zone in DSDP Hole 550, and the Oldhaven magnetozone all correspond to C24r.6. If this is true, then we would like to point out that, although this cryptochron occurs well above the Paleocene/Eocene boundary, nonetheless, it would be a useful stratigraphic marker in this time period of few reversals.

4.3. The Paleocene / Eocene boundary

Identification of the Paleocene/Eocene boundary remains a matter of much discussion [see 28]. There appears to be a general consensus that the nannofossil zonal boundary NP9/NP10 is a reasonable guide to its location in marine sediments, however. In Gulf Coast sections of the United States, there is a weak suggestion that this nannofossil boundary roughly corresponds with the FAD of *Platycarya* pollen [e.g., 9]. This form of pollen occurs in the Bighorn Basin composite section but its FAD is poorly constrained. Wing et al. [5] and Bown et al. [6] place it some 100 m above the base of the section, whereas [4] quote unpublished sources to place it at −34 m relative to the base of the section, as shown in Fig. 4. If the data and interpretations summarized in Fig. 5 are correct, then the Paleocene/Eocene boundary, as located in the first trimester of C24r, occurs within R1 of the Bighorn Basin composite section, near the horizon containing Wa0 faunas, as well as near to the lower placement of the FAD of *Platycarya* pollen. A more precise location cannot be resolved by the data in hand, partly because additional magnetostratigraphic data are required to document fully the lower part of the Bighorn Basin composite section and partly because no adequate, universally accepted, definition of the boundary exists.

4.4. Consequences for other continental sections

As noted previously, there is an independently studied sequence of sediments in the Clark's Fork Basin to the north (Fig. 1). (Please note that there is some confusion in the literature about names of the various sedimentary basins in Wyoming, whereby the Clark's Fork Basin is referred to as the Bighorn Basin. We retain the usage in Butler et al. [25]). These sediments are in part contemporaneous with those studied here [see 29 for a recent summary]; hence, the magnetostratigraphic results summarized in Figs. 4 and 5 have a bearing on interpretations of age relations in the Clark's Fork Basin. One of the consequences of the magnetostratigraphic correlation presented here is that the placement of the Paleocene/Eocene boundary at 160 m above the Wa0 faunal level in the southern Bighorn Basin by [29] is inconsistent with the constraints shown in Fig. 5. Thus, the change in body size of *Coryphodon* discussed by [29] probably post-dates the Paleocene/Eocene boundary. Identification of cryptochron C24r.6 in the northern Bighorn Basin would provide a more precise means of

correlation, however, and the details of the timing of mammalian responses to the climatic changes accompanying the Paleocene/Eocene boundary await higher resolution magnetostratigraphic results.

More recently, Koch et al. [30] have presented a 'correlation' of $\delta^{13}C$ data from deep sediments to those measured on pedogenic carbonate nodules in the sediments of the Clark's Fork Basin. There is an excursion in $\delta^{13}C$ data observed in the middle of C24r, as identified in ODP Hole 690B. There is a single-point 'excursion' in the δ^{13} data at the 1520 m level (Wa0) in the northern Bighorn Basin (see * in Fig. 5). No magnetostratigraphic data extend through Chron C24r. According to our correlation, however, the Wa0 level is likely to fall in the lower third of C24r — not the middle — and the 'correlation' of $\delta^{13}C$ data proposed by Koch et al. [30] is likely to be fortuitous.

Clarforkian/Wasatchian faunas have also been identified in the Big Bend National Park of Texas and placed in a magnetostratigraphic context by Rapp et al. [31]. Their fossil locality at 137 m was approximately in the middle of a reversed zone confidently correlated to C24r. The mammalian zone was tentatively identified as Clarkforkian, although the authors describe several mammals that are usually assigned to the Wasatchian land mammal age (e.g. the FAD of *Hyracotherium*). We see from Fig. 5 that the lowest unit of the Wasatchian land mammal age (Wa0) occurs some 100 m below cryptochron C24n.6, or roughly one-third of the way through C24r. Thus, the 137 m locality of Rapp et al. [31] is likely to be close to the Clarkforkian/Wasatchian boundary (if not actually above it). The identification of C24n.6 in the Big Bend section would facilitate high-resolution correlation, but the sampling density of [31] was insufficient to detect such a short feature of the geomagnetic field.

4.5. Sediment accumulation rates

Based on the interpretation summarized in Fig. 5, the composite section in the Bighorn Basin spans from about 55 Ma (near the Paleocene/Eocene boundary) to about 52.5 Ma at the top of the Willwood Formation (about 690 m). Thus, the average sediment accumulation rate was about 290 m/m.y. During C24n.3, however, the average accumulation rate was higher, about 430 m/m.y. This is consistent with the interpretation of Bown and Kraus [19], who supposed higher sedimentation rates at the top of the section, based on maturation of the paleosols. Since maturation of soils is a function of climate as well as of time, the independent age information provided by the magnetostratigraphy could provide a powerful test of the pedogenic model. For example, a lateral magnetostratigraphic study (similar to that presented by [32]), spanning one of the magnetozones documented here and including several pedogenic styles, would provide crucial independent control on sediment accumulation rates.

4.6. Calibration of the GRTS

Calibration of the GRTS in terms of numerical age relies principally on tying isotopic dates to the reversal pattern [see 1 for a recent discussion of the problem]. There are very few reliable dates so correlated within the Paleogene, hence the large discrepancies in age estimates for the Paleocene/Eocene boundary. Harland et al. [2] discuss the isotopic age estimate of 53 Ma for the boundary, based on glauconite data, and suggest that this may be too young. In their 'definitive' time scale calibration, they favor an older age of 56.8 Ma. They estimate the age of the older boundary of C24n.1 as being 53.69 Ma. In an alternate calibration, Cande and Kent [1] quote an age of 55 Ma (due to Swisher and Knox, unpublished) for the Paleocene/Eocene boundary and derive an age for the older boundary of C24n.1 of 52.54 Ma.

We sampled through the section that contained the tuff dated by [4] at 52.8 ± 0.3 Ma. The tuff lies between sediments of the R6/N6 polarity zones, which we correlate to the older boundary of C24n.1. Thus, according to our interpretation, the tuff provides the best estimate for the age of the boundary, which is within the uncertainty, indistinguishable from that of [1]. Thus, not only do we concur with [2] in rejecting an age as young as 53 Ma for the Paleocene/Eocene

boundary, we find the time scale of Cande and Kent [1] provides a more accurate age for Chron C24, than that of [2].

5. Summary and conclusions

The paleomagnetic behavior of the sediments in the Bighorn Basin composite section, although complicated, can be interpreted in terms of a magnetostratigraphic polarity zonation. Correlation of the polarity zones to the GRTS of Cande and Kent [1] is facilitated by an isotopic date of 52.8 ± 0.3 Ma from a tuff near the top [4] and by including the magnetostratigraphic data of Butler et al. [25] from the Clark's Fork Basin to the north. The magnetostratigraphy can thereby be correlated with reasonable confidence to Chrons C24n.1–C24r. We also tentatively correlate a small magnetozone near the base of the section with cryptochron C24r.6. If we are correct, C24r.6 represents a full reversal of the geomagnetic field as opposed to a fluctuation in intensity.

The highest resolution magnetostratigraphic data from marine sediments through C24r were obtained from DSDP hole 550 [23]. The Paleocene/Eocene boundary (as recognized by nannofossil zonal boundary NP9/NP10) occurs within the first trimester of C24r [24]. There is a normal zone, defined by a single positive inclination, at about the mid-point of C24r in DSDP 550. This can be tentatively correlated with a normal zone (N1?) in our section and with Cryptochron C24r.6 of the geomagnetic reversal time scale [1]. If this correlation is valid, then the Paleocene/Eocene boundary occurs near the base of the composite section in the Bighorn Basin, or close to the local level of the Clarkforkian/Wasatchian land mammal age boundary. The FAD of *Platycarya* pollen also occurs near the base of the section. This pollen form is loosely correlated to the NP9/NP10 boundary in the Gulf Coast of North America [9]. A more precise location of the Paleocene/Eocene boundary depends critically on its definition, which has not yet been formalized.

The magnetostratigraphic interpretation put forward here has consequences for sections in the northern Bighorn Basin. On the basis of correlating basal Wasatchian faunas between the Bighorn Basin and the Clark's Fork Basin in Wyoming, we suspect that the identification of the Paleocene/Eocene boundary in the work of [29] to be incorrect. Hence, their interpretation of a change in body size in *Coryphodon* as a reflection of the climatic change accompanying that Paleocene/Eocene boundary is poorly established. Moreover, the supposed connection between $\delta^{13}C$ data from ODP Hole 690B and from the Clark's Fork Basin is not supported by our data. However, recognition of C24r.6 in sediments of the Clark's Fork Basin would provide a high-resolution link between the continental and the marine records.

Our data support the supposition of Bown and Kraus [19], based on soil maturation profiles, of higher sediment accumulation rates in the upper part of the Willwood composite section. As pedogenic processes are a function not only of time but of climate, an independent chronological framework based on parallel magnetostratigraphic sections could provide a crucial test for models of soil maturation rates.

Finally, the correlation of the isotopically dated tuff (52.8 ± 0.3 Ma) to the older boundary of C24n.1 suggests that the date of 52.5 Ma estimated by Cande and Kent [1] is preferable to the date of 53.7 Ma estimated by Harland et al. [2]. Our data essentially rule out an age as young as 53 Ma for the Paleocene/Eocene boundary.

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